



LINSON PHILIP

Electrical & Energy Data Engineer

Power Systems | Energy Meter Testing | Test Lab Automation

Phone +49 152 10726235	Email linsonphilip5@gmail.com	LinkedIn linkedin.com/in/linson-philip
Portfolio linsphilip.github.io/linson-portfolio	Location Ortenberg, Germany	Mobility Driving licence B197

Professional Profile

Electrical and energy-data engineer with experience in EHV power-system operations, energy-meter testing, measurement-data analysis, digital-twin development and laboratory automation. Currently working on large-scale meter-test data analysis and automated three-phase test-system integration at Hochschule Offenburg.

Professional Experience

Academic Research Associate 05/2025–present
Institute for Sustainable Energy Systems (INES), Hochschule Offenburg

- Analyse more than **2 million historical energy-meter test records** using Python, Pandas and SQL Server to assess measurement error, long-term drift and device-level stability.
- Develop data pipelines for test-point harmonisation, validation, KPI generation and automated technical reporting.
- Design and integrate an **automated three-phase energy-meter test laboratory** combining programmable power equipment, electronic loads, reference measurement, tariff meters and climate-testing equipment.
- Develop equipment-control and data-acquisition workflows using Python, Node-RED, Modbus TCP/RTU, RS-485, GPIB/VISA, MySQL and InfluxDB.

Master’s Thesis Student - Digital Twin Development 09/2024–02/2025
Livarsa GmbH & Hochschule Offenburg

- Developed and validated a digital twin to assess operating behaviour and energy-saving potential of an industrial electrical efficiency filter system.
- Performed electrical measurements and parameter identification; modelled the system with LTspice and MATLAB/Simscape using real measurement data.

Student Research Assistant - Micro-PV Monitoring 12/2023–09/2024
Institute for Sustainable Energy Systems, Hochschule Offenburg

- Supported installation and monitoring of micro-PV systems and evaluated operational energy data.
- Developed Python-based workflows for data preparation and technical analysis.

Electrical Engineer - EHV Substation Operations 09/2018–10/2021
Kerala State Electricity Board, Kerala, India

- Operated and monitored 11–110 kV substations and transmission equipment, including switching operations and operational documentation.
- Worked with power transformers, circuit breakers, isolators, protection relays, batteries and DC auxiliary systems; supported insulation and condition testing.
- Coordinated maintenance, planned shutdowns and restoration activities while monitoring voltage, current, loading and energy flow.

Technical Skills

Power & Measurement	Power systems, energy-meter testing, electrical measurements, three-phase systems, power quality, EHV substations, transformers, switchgear, protection systems
Data & Programming	Python, SQL, Pandas, NumPy, Scikit-learn, statistical analysis, time-series analysis, KPI development
Automation	Node-RED, data acquisition, automated test sequences, equipment integration, MySQL, InfluxDB
Communication	Modbus TCP/RTU, RS-485, GPIB, VISA, Ethernet/TCP-IP
Modelling	MATLAB, Simulink, Simscape Electrical, LTspice, digital twins, parameter identification

Education

M.Sc. Renewable Energy and Data Engineering
Hochschule Offenburg, Germany 2023–2025

B.Tech. Electrical and Electronics Engineering
University of Kerala, India 2014–2018

Selected Projects

Energy-Meter Accuracy Assessment 2025–present

Historical meter-test data analysis for measurement error, long-term drift and stability assessment; KPI and reporting workflows using Python and SQL.

Automated 3-Phase Test Laboratory 2026–present

Integration of programmable power equipment, electronic loads, reference measurement, energy meters and databases using Node-RED, Modbus, RS-485 and GPIB/VISA.

Digital Twin - Efficiency Filter 2024–2025

Measurement-based parameter identification and development/validation of LTspice and MATLAB/Simscape models.

Training

Python for Data Science
DataCamp

Languages

Malayalam - Native	English - C1
German - B2	Tamil - B1
Hindi - B1	

Additional Information

Driving Licence
German driving licence, Class B197

Location
Ortenberg, Germany